This procedure is applicable to below models:

V810 STD		☐	S1	☐	S2
V810 XXL		☐	S1	☐	S2
V810 S2EX		√			
V810 Mini		☐			

REVISION HISTORY

Rev	Effective Date	Description	Doc Originator
00	22 May 2017	First Release	CHIN CHEE YAN

Content

Removal Procedure

1. **Shutdown and completely power off machine**. Turn off the main air pressure regulator
2. **Power off the SSCA main power** # Importance.
3. Open front and rear access door
4. Loosen 2 x Tamper Proof screws as below Figure 1. Gently open outer barrier

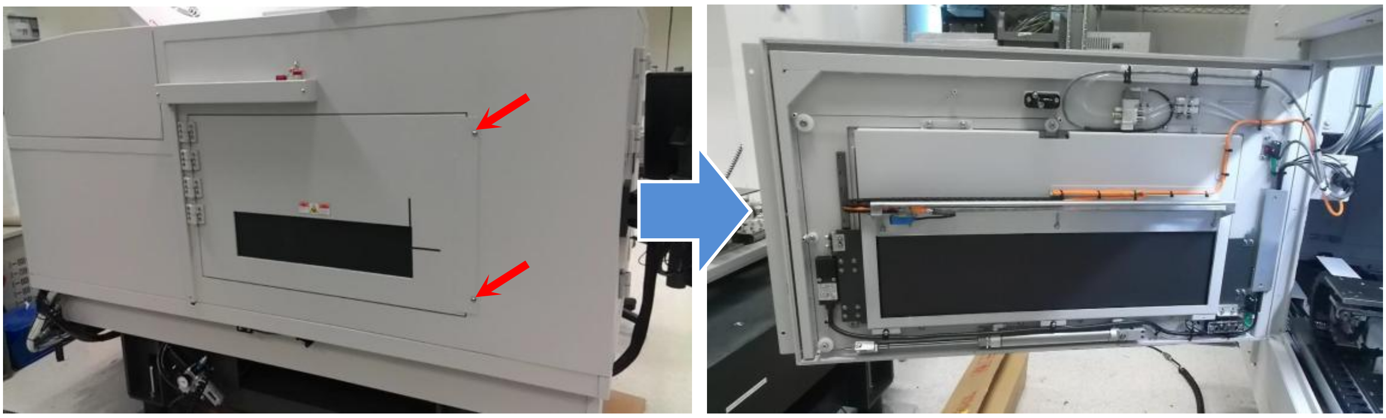


Figure 1 Left Outer Barrier (Open)

5. Manually move the XY stag towards left outer barrier
6. Disconnect the Cable, Motor Stepper AWA Axis Adjustable Rail (P/N, 4JASC-A078) from Left AWA motor. (Remarks : Left Motor location define by Front Door Direction)

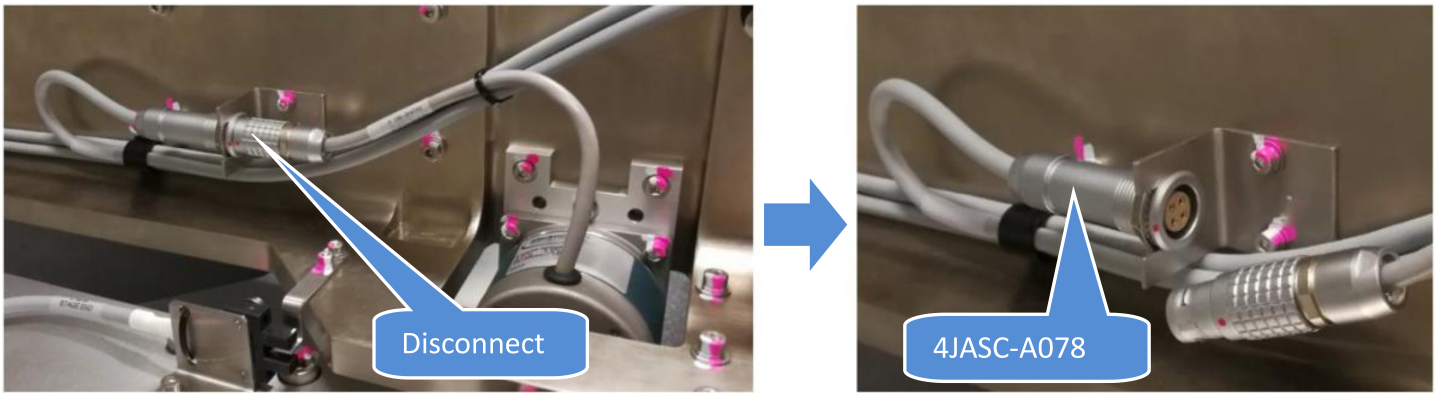
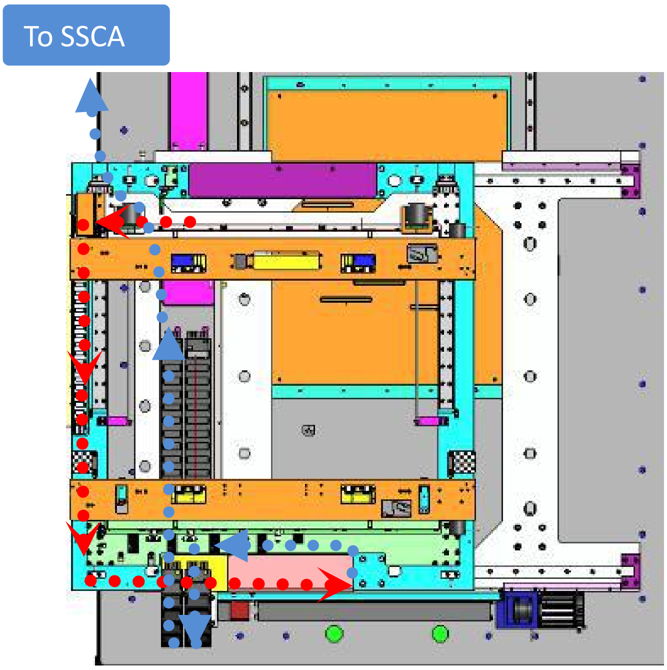


Figure 2 Left Stepper Motor Connector Removal

7. May refer to below diagram for Left Stepper Motor Cable Routing

Route : Left Stepper Motor -> AWA E-Chain -> X axis E-Chain -> Y axis Wide E-Chain -> SSCA



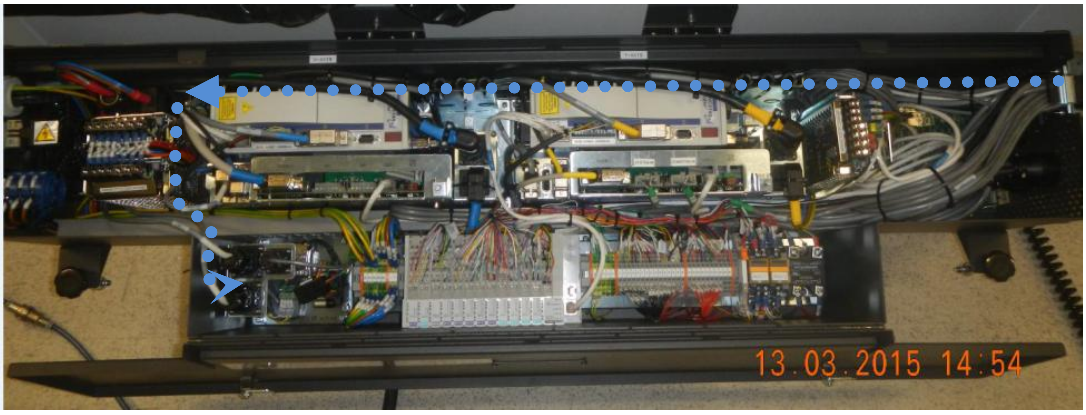


Figure 3 Left Stepper Motor Cable Routing

8. Loosen the lock nut at Cable, Motor Stepper, AWA Axis (4JASC-A078) from BRACKET, SINGLE PKG (P/N, 4XAXY-061) by using 17mm wrench.

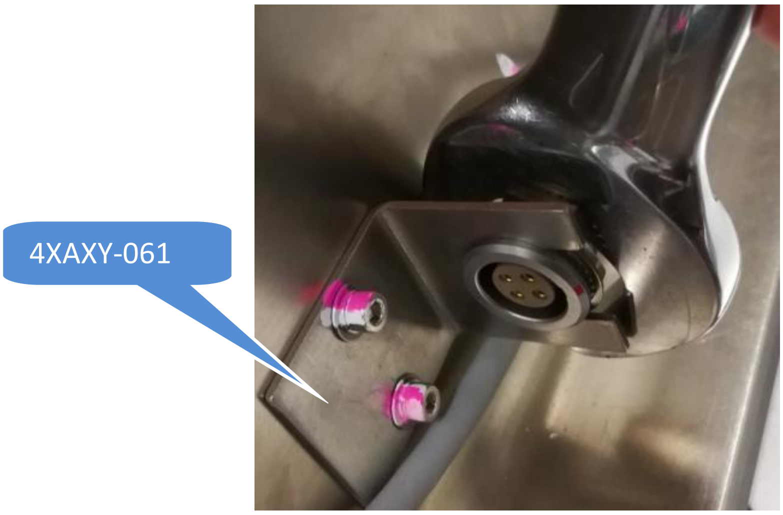


Figure 4 Lock nut Loosen

9. At the adjustable rail, loosen Cable Clamp and remove 4JASC-A078 from Cable Clamp. Remove cable tie as well

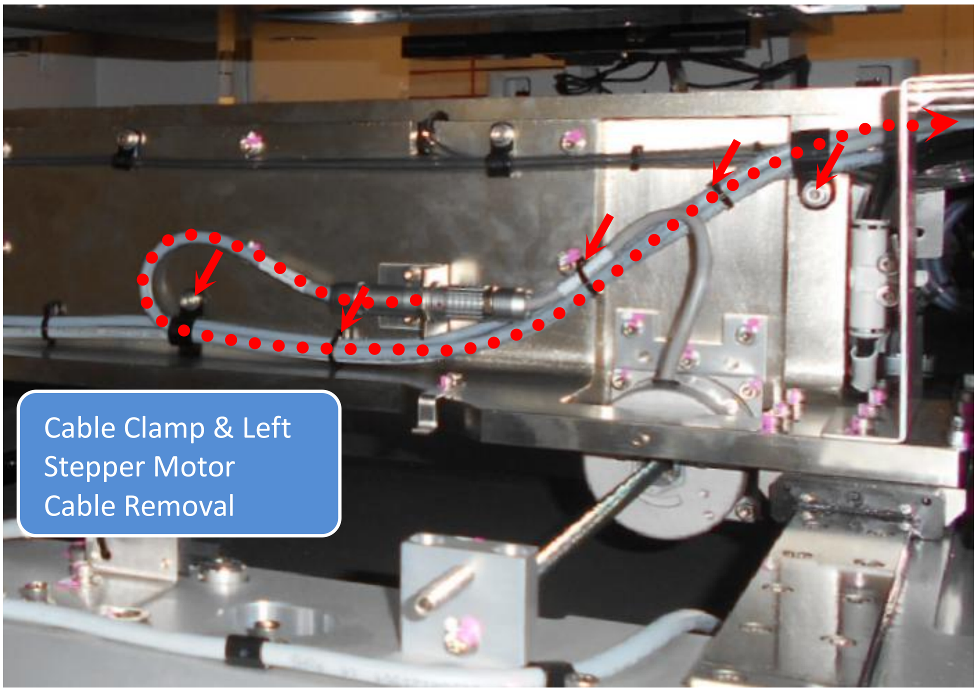


Figure 5 Cable Clamp & Left Stepper Motor Cable Removal

10. Loosen 4 x Screws and remove CHAIN BRACKET (P/N, 4JAXY-036) as below Figure 6

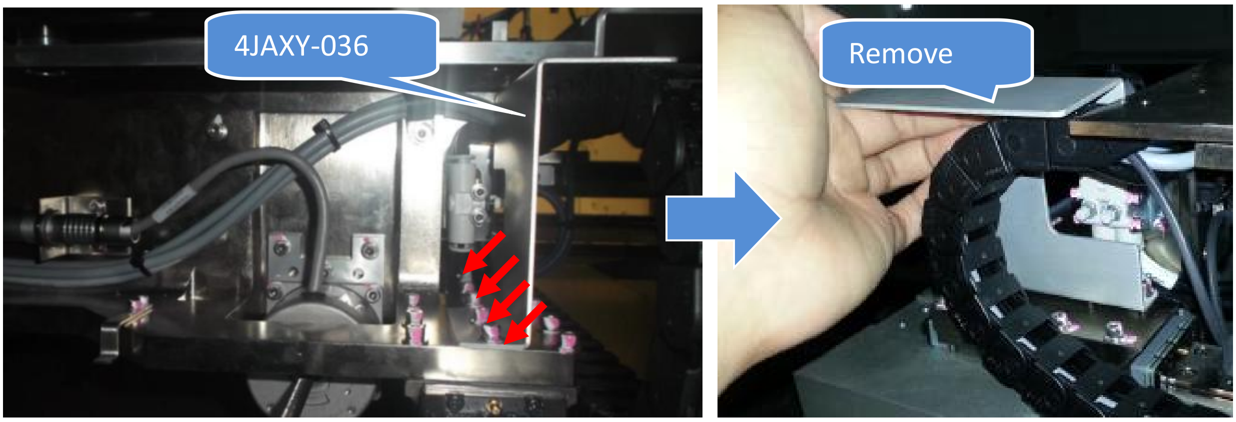


Figure 6 Chain Bracket Removal

11. Use flat head screwdriver to open AWA E-Chain Top Joint

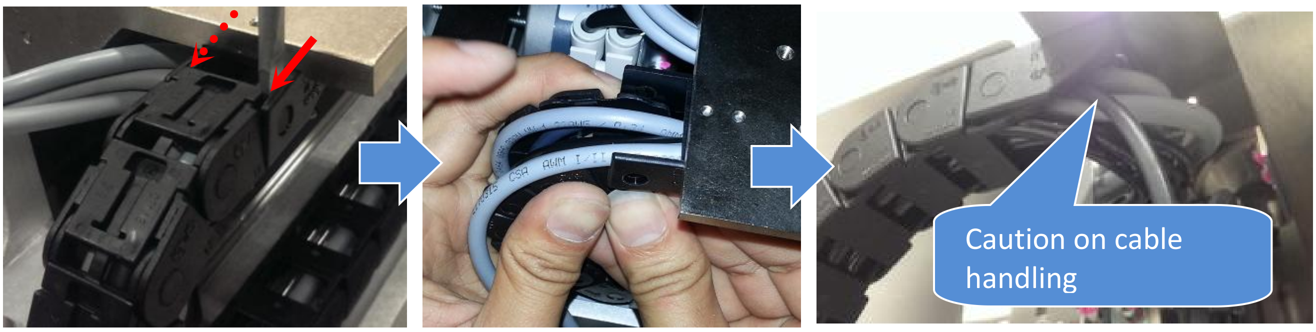


Figure 7 AWA Top Joint Removal

12. Use flat head screwdriver to open Slot Cover along the AWA E-Chain

Figure 8 E-Chain Slot Cover Open

13. Use flat head screwdriver to remove AWA E-Chain bottom joint as below Figure 9

Figure 9 E-Chain Bottom Joint Removal

14. Remove 4JASC-A078 from AWA E-Chain as below Figure 10

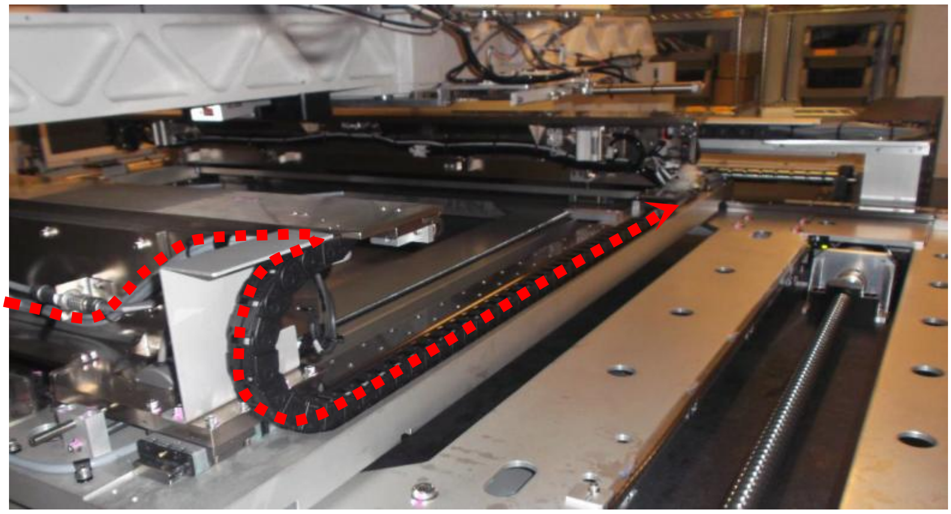


Figure 10 Stepper Motor Cable Removal (AWA E-Chain)

15. At Fix Rail, loosen 3 x screws and remove the PIP Cable Cover (P/N , 4JAXYS-N104) as below Figure 11

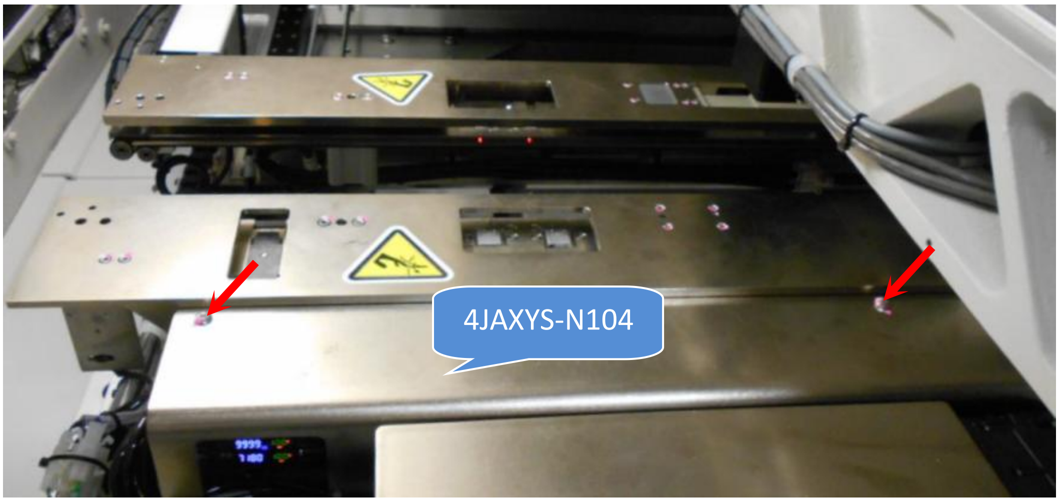


Figure 11

16. Gently lift up Motor Cables and loosen 2 x screws as shown.Open & remove the GUARD, E-CHAIN (P/N, 4JAXY-047).

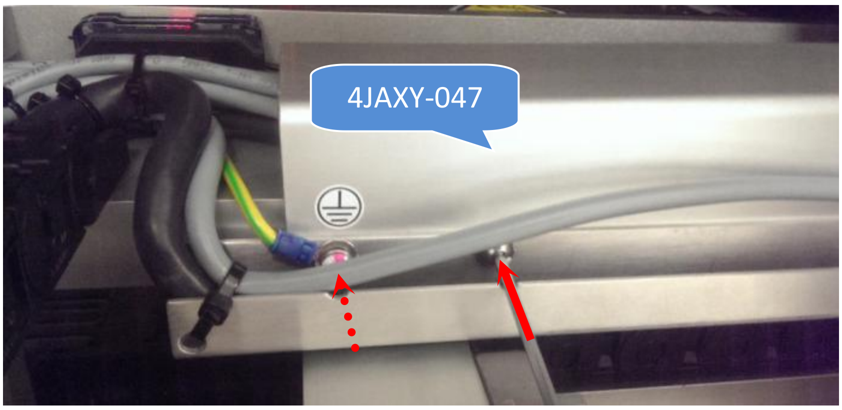


Figure 12 X Axis E-Chain Guard Plate

17. Track remaining route and remove stepper motor cable. Remove necessary cable clamp & cable tie

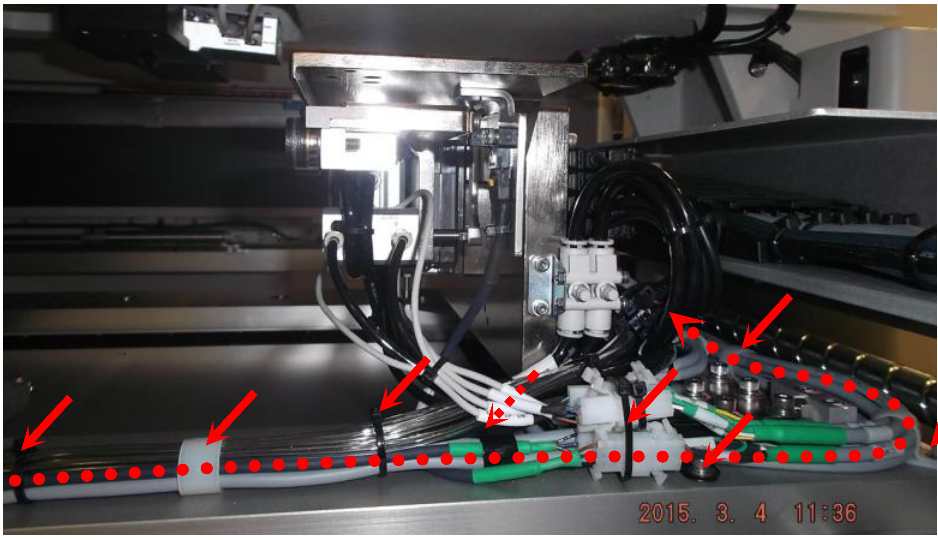


Figure 13

18. Use flat head screwdriver to open V810 S2EX X-AXIS E-CHAIN (P/N, 4JAXYS-A117) end joint (top and bottom) as below Figure 14

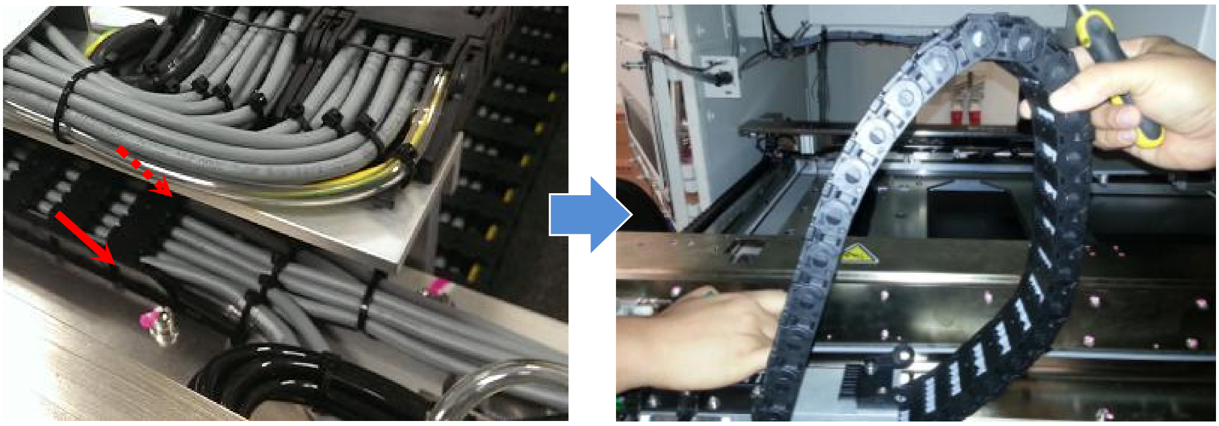


Figure 14 X Axis End Joint Removal

19. Use flat head screwdriver & open E-chain cover top cover along X axis E-Chain

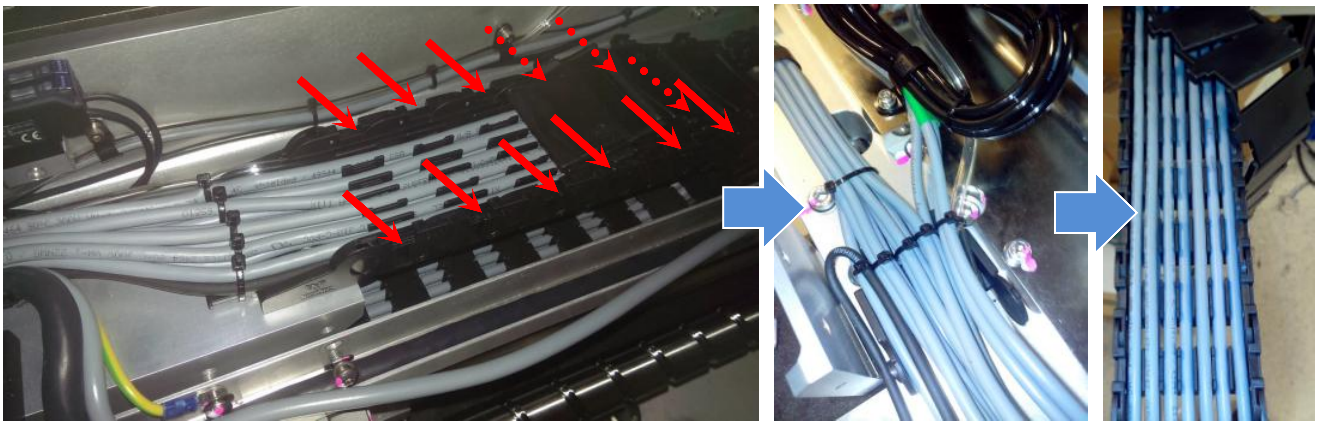


Figure 15 X Axis E-Chain Cover Removal

20. Remove 4JASC-A078 from X axis E-Chain as below Figure 16. Remove necessary cable tie

Figure 16 Stepper Motor Cable Removal (X Axis E-Chain) -Reference Picture Only

21. Track 4JASC-A078 cable route along Y axis Wide E-Chain.Use flat head screw driver to remove E-Chain Top cover along Y axis Wide E-Chain. Next,open Y axis Wide E-Chain bottom joint

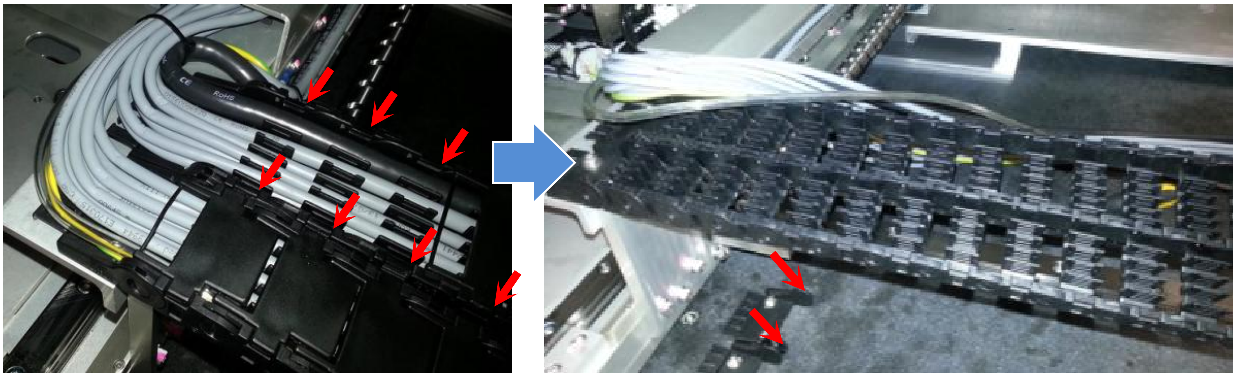


Figure 17 Stepper Motor Cable (Y Axis Wide E-Chain)-Reference Picture

22. Remove 4JASC-A078 from Y axis Wide E-Chain. Remove necessary cable tie as well

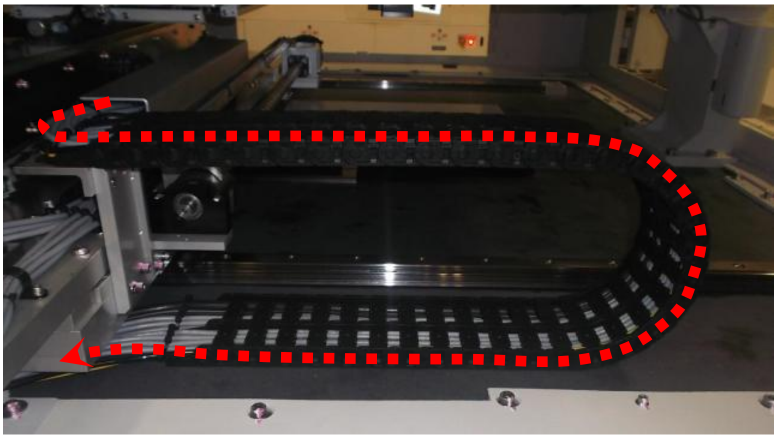


Figure 18 Stepper Motor Cable Removal (Y axis Wide E-Chain)

23. Track remaining route and remove 4JASC-A078 accordingly. Remove necessary cable tie.

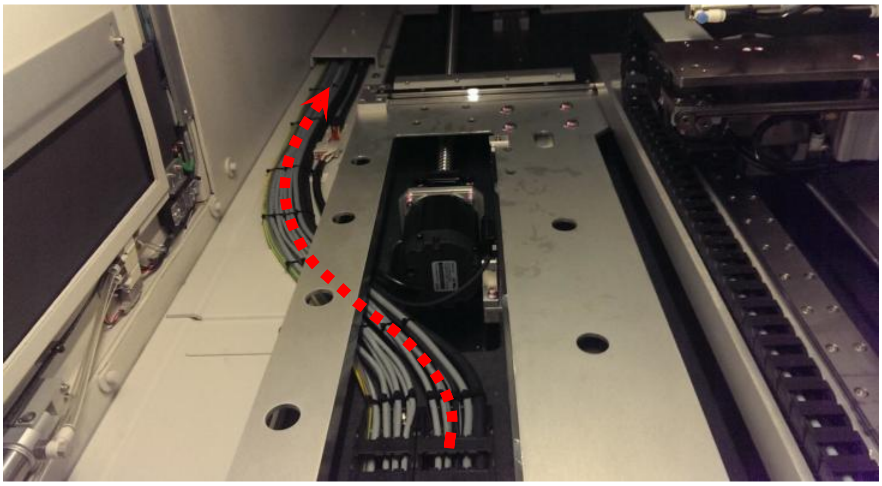


Figure 19 Stepper Motor Cable Removal

24. Open Rear side door & loosen 5 x Screws.Remove" Vertical Lead Shielding Assembly" and "Cable Exit Shielding Assembly" as below Figure 20

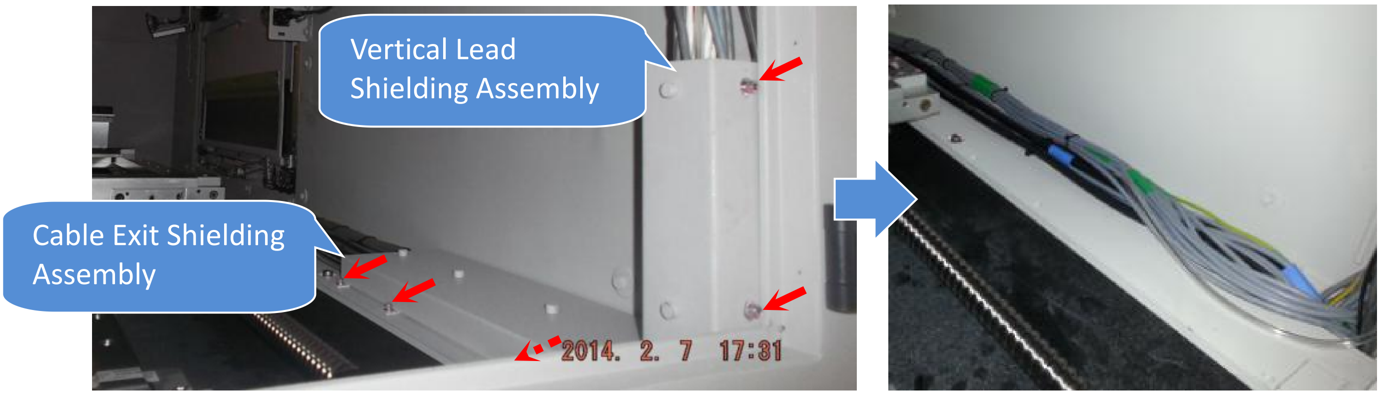


Figure 20 Shielding Assembly Removal

25. Gently remove 4JASC-A078 through the "Opening"

Figure 21

26. Gently remove 4JASC-A078 as shown. Remove necessary cable tie

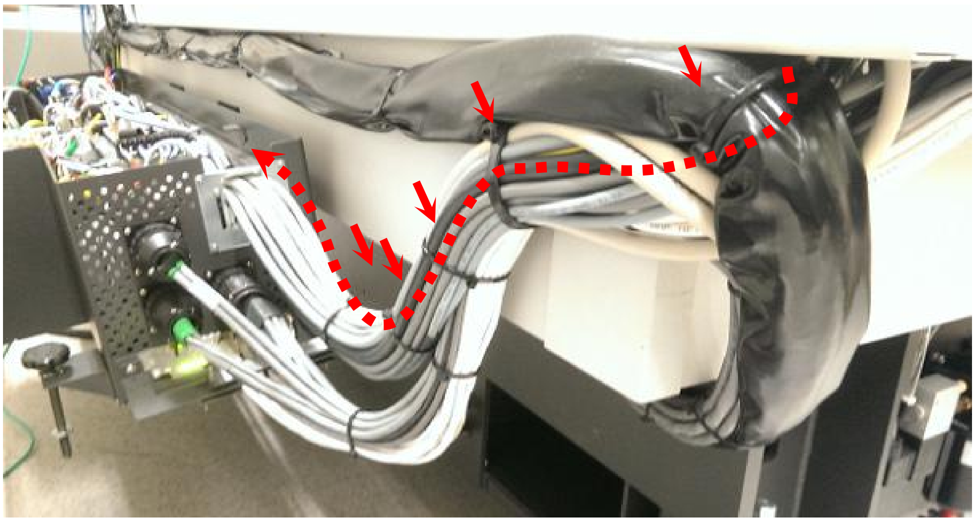


Figure 22 Stepper Motor Cable Removal

27. Loosen 2 x Screws.Remove MCA Lambda HWS 100 Body Holding Plate Assembly (P/N, 4XAMC-A045) as shown.

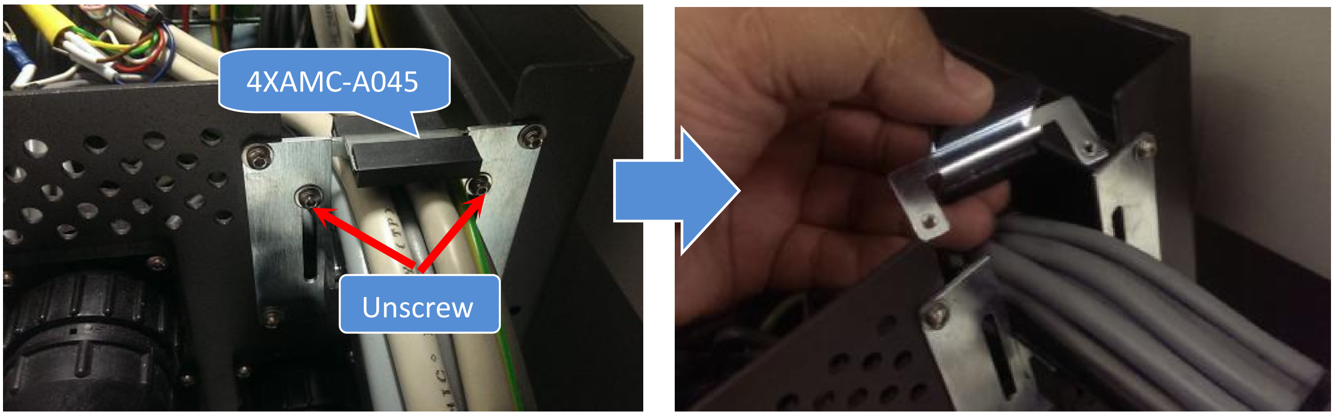


Figure 23 Holding Plate Removal

28. Track 4JASC-A078 route at SSCA.

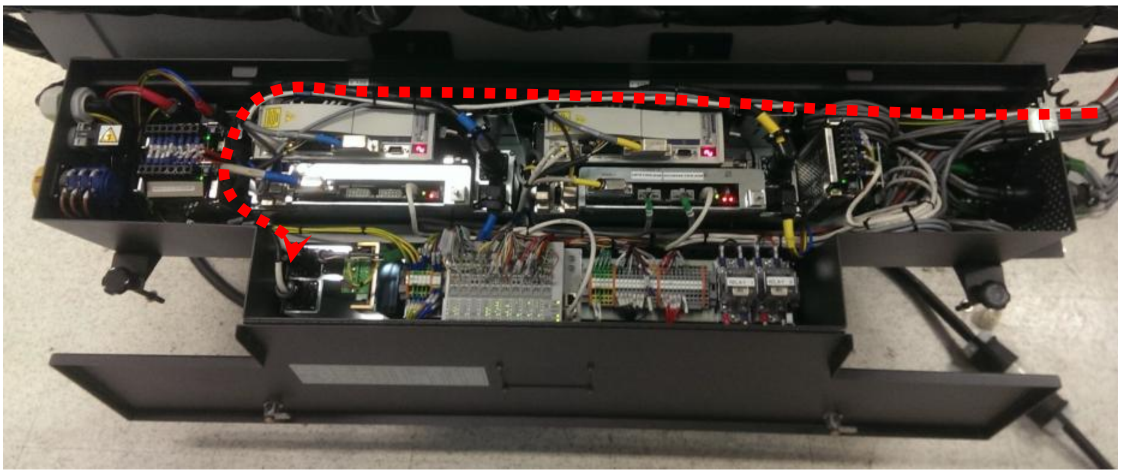


Figure 24 Stepper Motor Cable Removal (SSCA)

29. Track stepper motor cable end. Disconnect bulb head connector by refer to Cable Label and remove from stepper driver

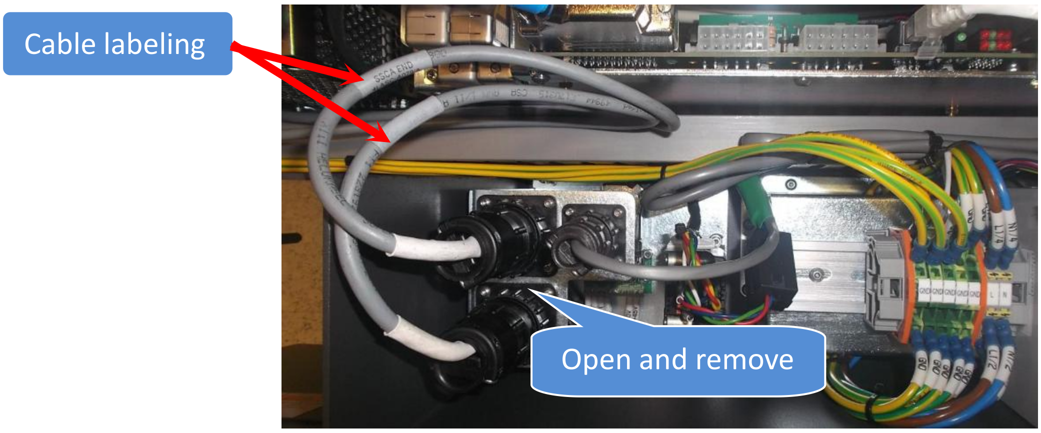
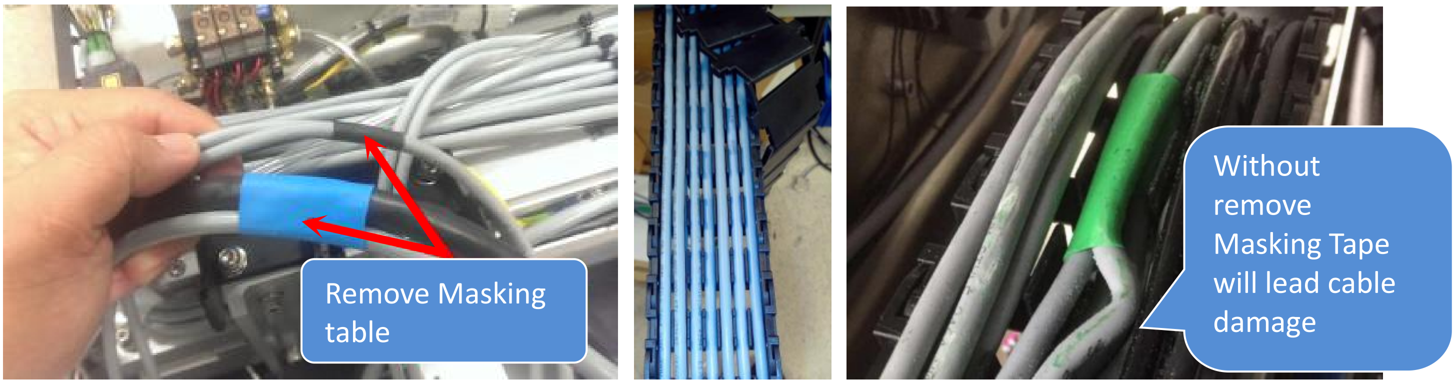


Figure 25 Stepper Motor Cable Removal (Stepper Driver)

Replacement Procedure

Caution : Cable within E-Chain where apply with Masking tape MUST REMOVE



Spare Cable Masking tape | Good routing | Bad routing

Figure 26

30. Prepare new CABLE, MOTOR STEPPER AWA AXIS (P/N, 4JASC-A078) as shown.

Figure 27

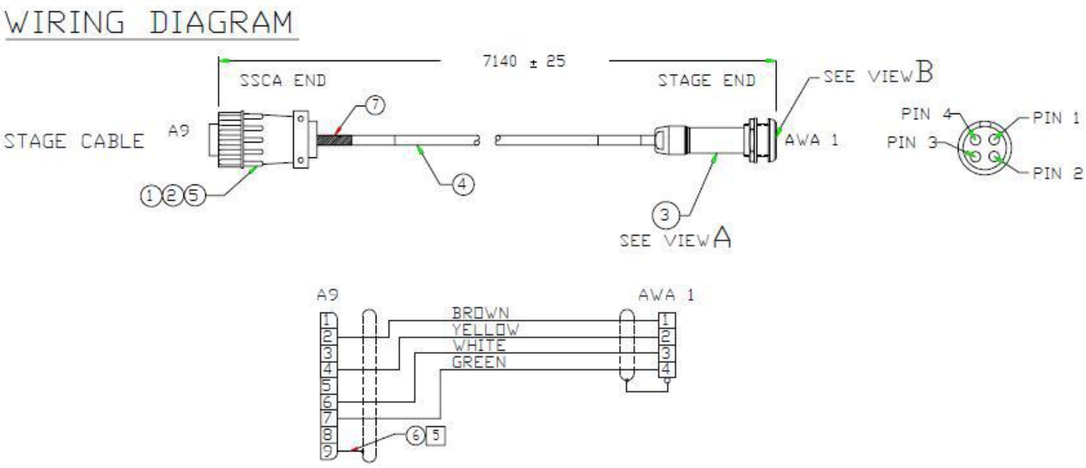
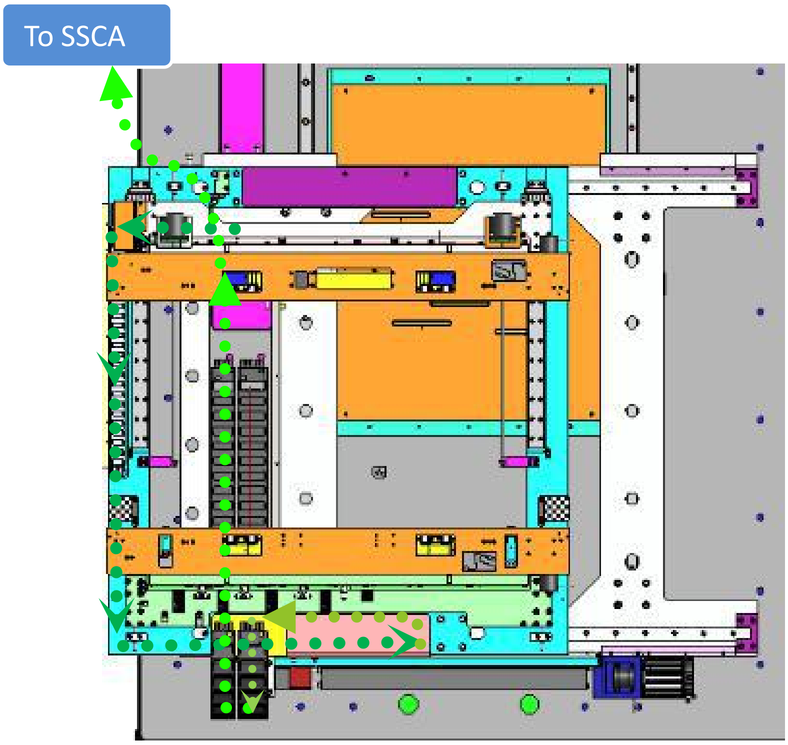


Figure 28 Cable Drawing

31. May refer to below diagram for Left Stepper Motor Cable Routing.

Route : Left Stepper Motor **->** AWA E-Chain -> X axis E-Chain -> Y axis Wide E-Chain -> SSCA



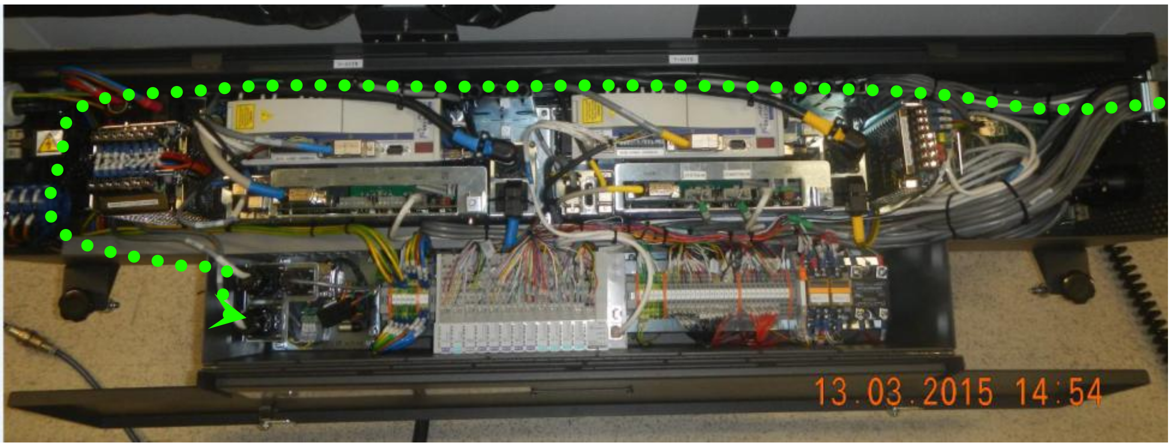


Figure 29 Stepper Motor Cable Routing

32. Assemble Stepper Motor Cable to holding bracket by refer to below Red Dot Orientation .Tighten lock nut by using 17 mm wrench.

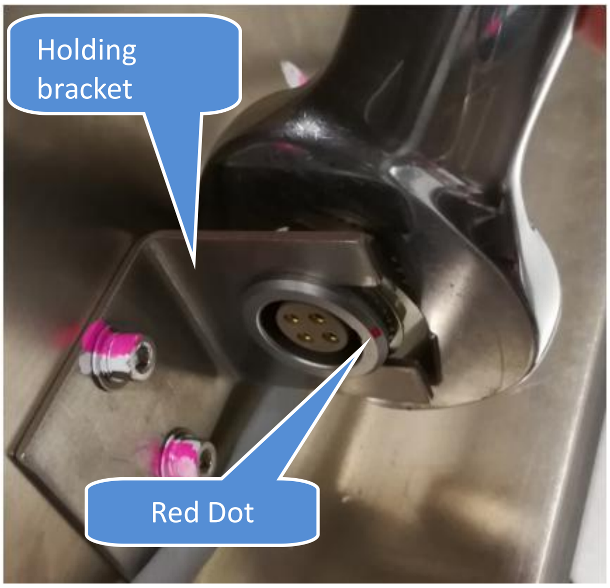


Figure 30 Stepper Motor Cable Assembly

33. Gently route the Stepper Motor Cable as below Figure 31. Secure the cable with Cable Clamp as well.

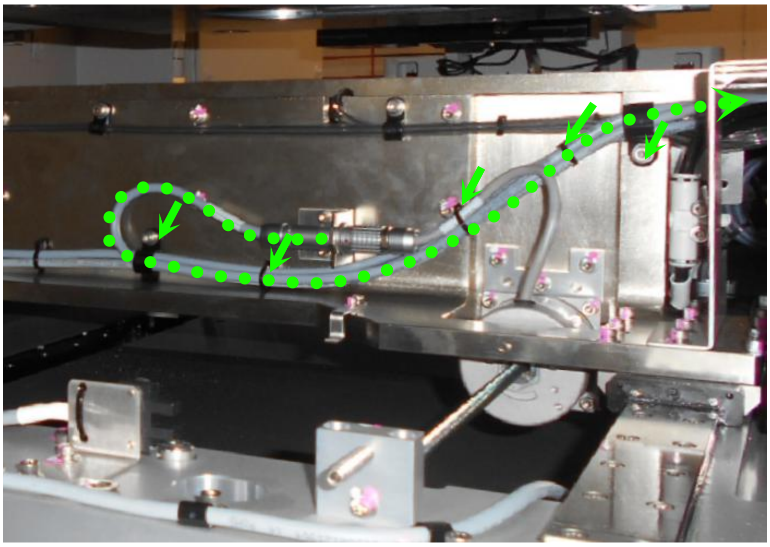


Figure 31 Stepper Motor Cable Routing

34. Connect AWA Stepper motor with 4JASC-A078 respectively by refer to Red Dot Orientation

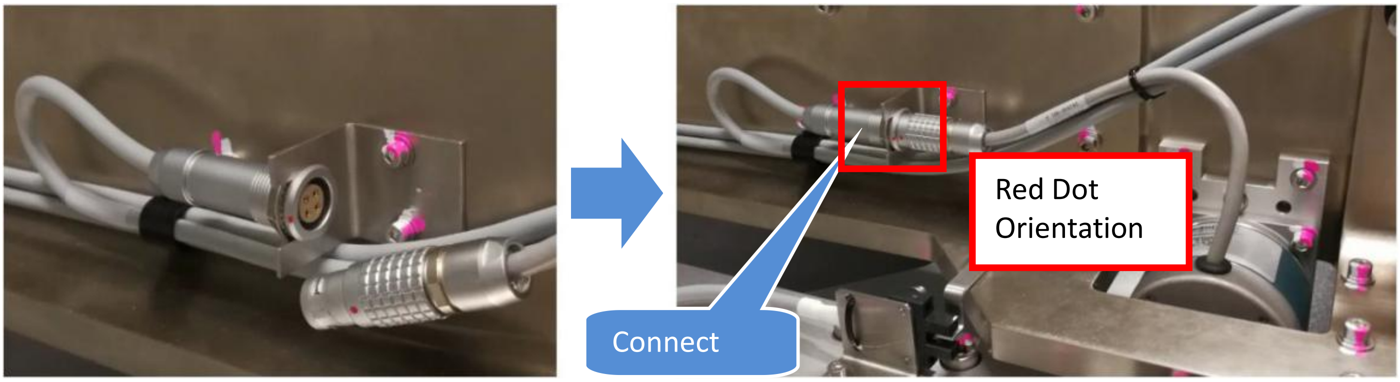


Figure 32 Stepper Motor Cable Connect

35. Route the cable with U shape curve into E-Chain as shown

    Remarks : Minor tolerance will require to apply to prevent cable snapping

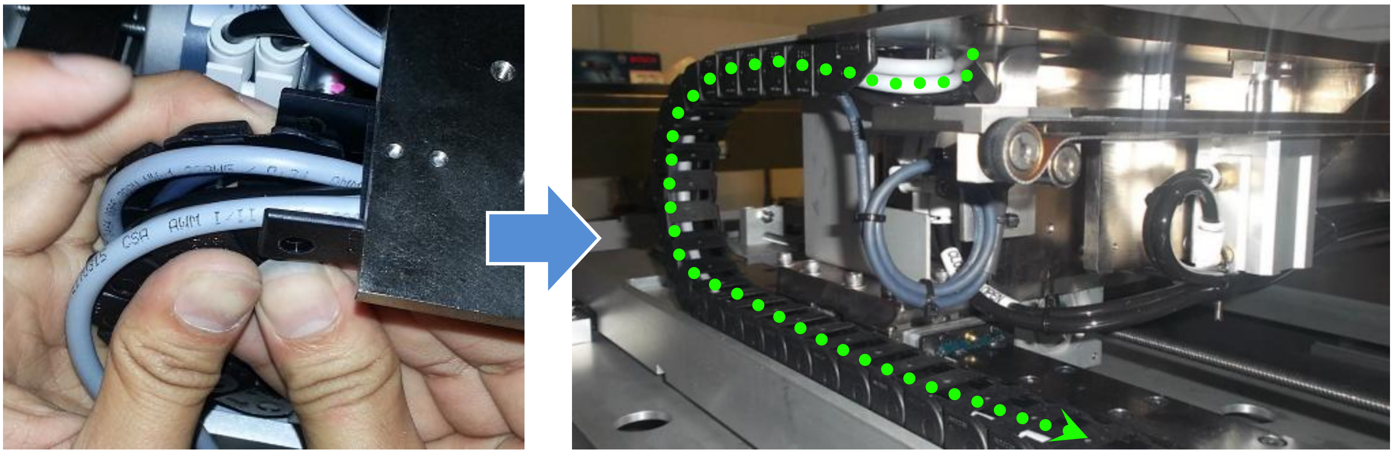


Figure 33 Stepper Motor Cable Routing (AWA E-Chain)

36. Assemble E-chain Top cover at AWA E-Chain. Next,reconnect the E-Chain Top Joint. Secure the AWA Cable with cable tie as below Figure 34

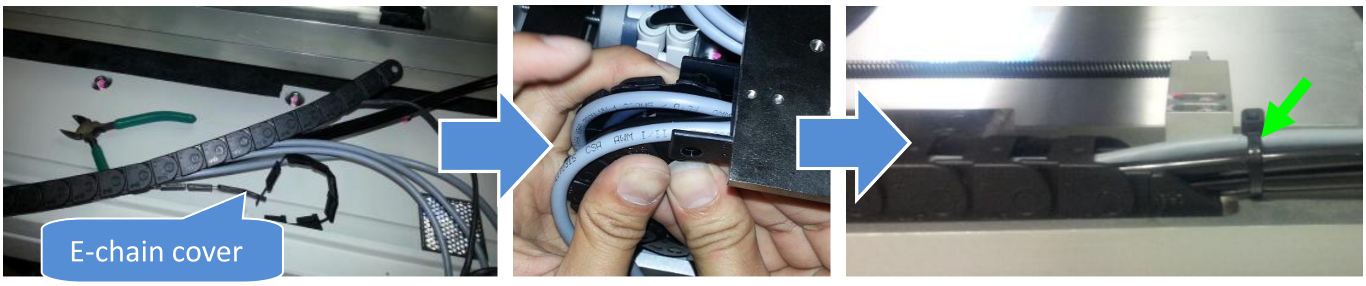


Figure 34 Cable Routing (AWA E-Chain)

37. Assemble 4JAXY-036 to Adj Rail. Tighten 4 x Screws accordingly

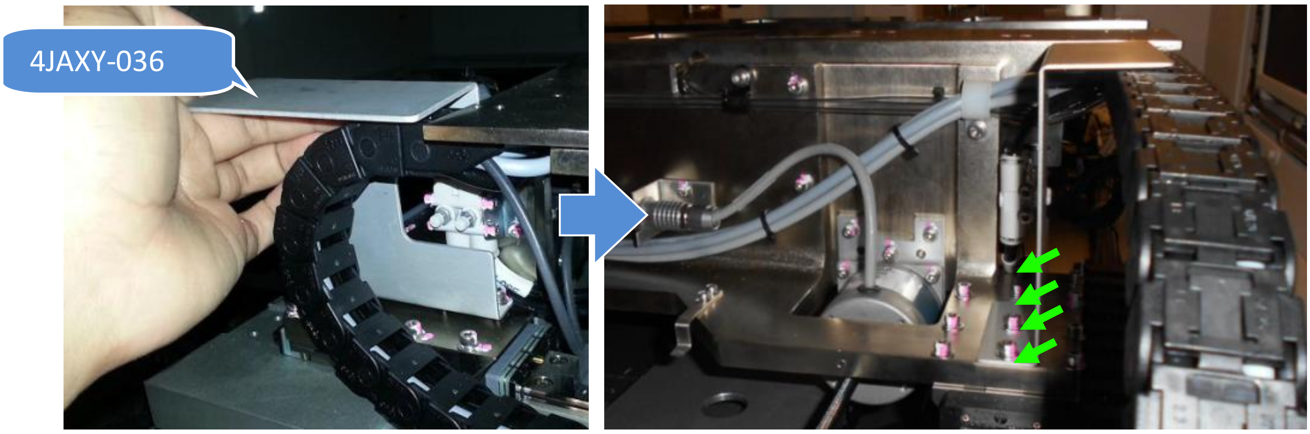


Figure 35 E-Chain Bracket Assembly

38. Gently route the cable as below Figure 36.Secure cable with Cable Clamp & Cable Tie.

Remarks : Caution on Set Screws which protrude at Fix Rail

Figure 36 Stepper Motor Cable Routing (Fix Rail)

39. Gently route Stepper Motor Cable into X axis E-Chain by refer to below criteria :

- Individual separation layout
- Cable do not get caught each other in the shelf parts
- Masking Tape MUST remove

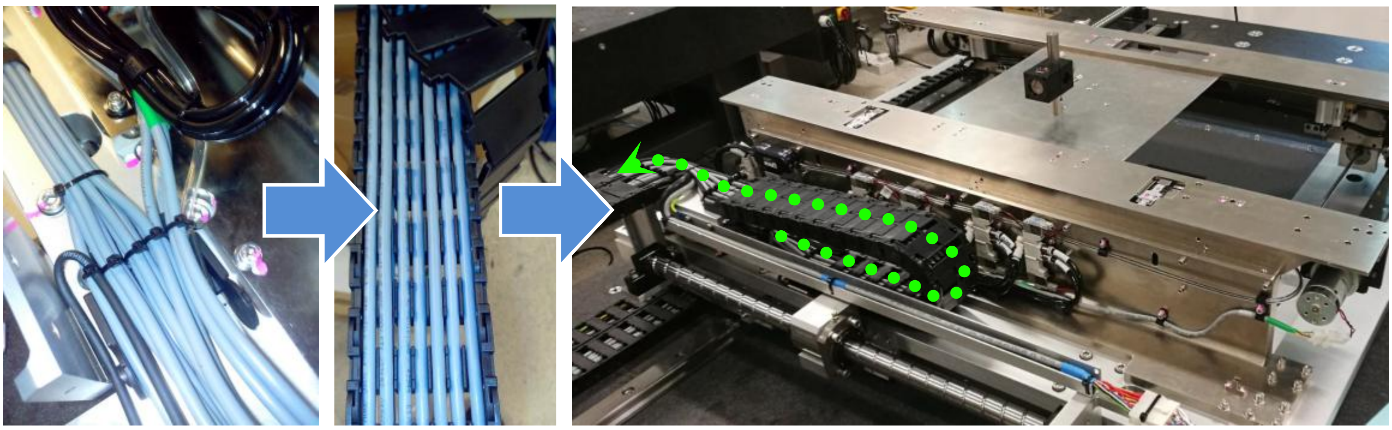


Figure 37 Stepper Motor Cable Routing (X Axis E-Chain)-Reference Picture Only

40. Gently push BRACKET, ENCODER CABLE (P/N, 4JAXY-076) towards rear side .Tighten 2x screws, & reassemble E-Chain cover. Secure Cable with cable tie as below Figure 38

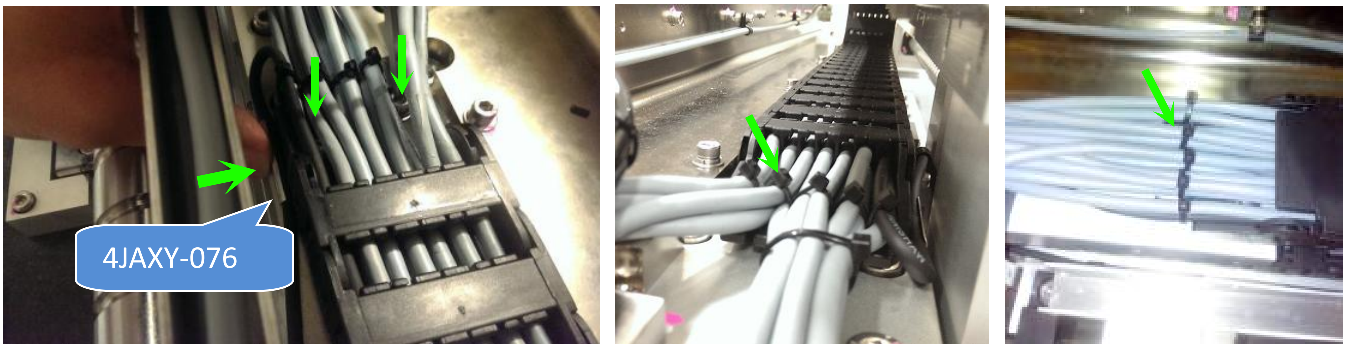


Figure 38 Stepper Motor Cable Routing (X Axis E-Chain)

41. Gently move X axis stage and visual check below item :

- X axis Encoder cable do not contain friction with encoder bracket
- Valve Assembly (Cable & Tubing) do not contain friction with X Axis E-Chain

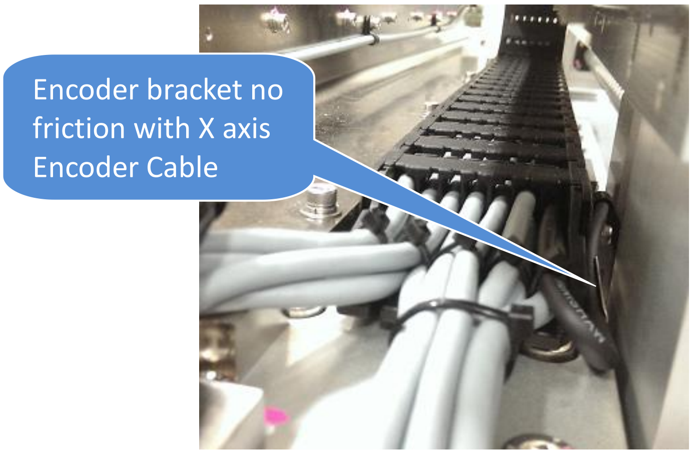


Figure 39

42. Manually move X axis stage towards left & right rapidly and visual check X axis E-chain where do not contain excessive bending radius

Remarks : Re-route the cable if excessive bending radius addressed

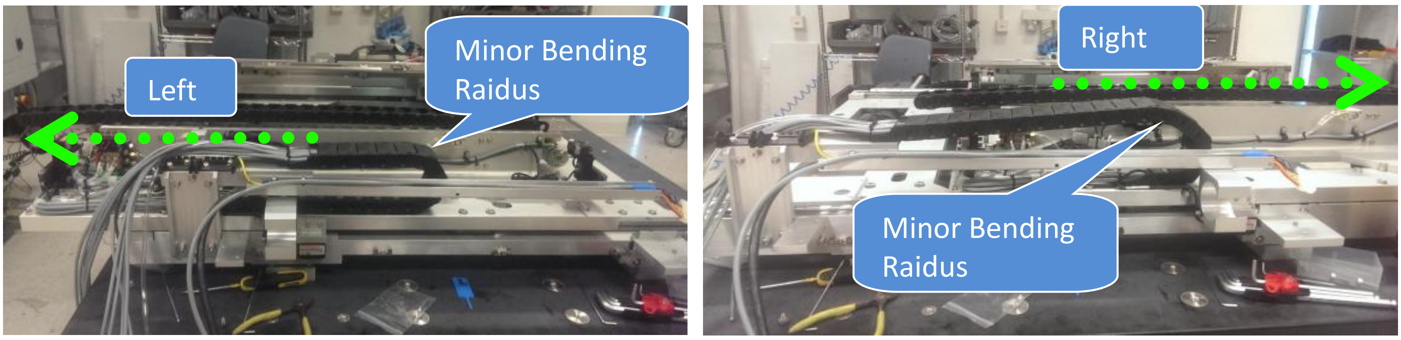


Figure 40 Minor Bending Radius (X axis E-Chain)

43. Assemble 4JAXY-047 as below Figure 41. Secure cables with cable tie

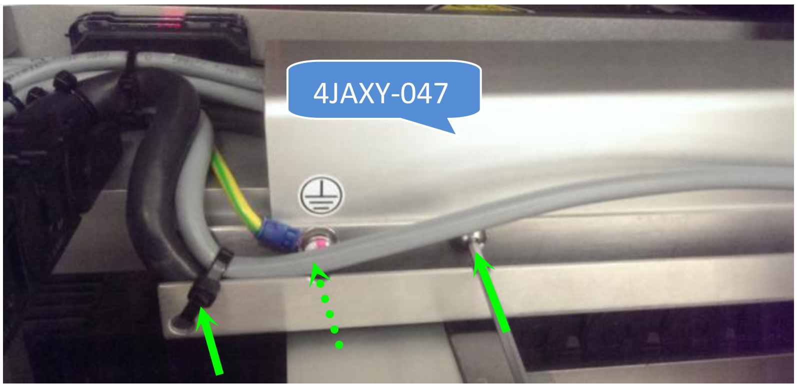


Figure 41 X Axis Guard Plate assembly

44. Route Stepper Motor cable at Y axis Wide E-Chain. Reassemble E-Chain cover.Connect the joint and secure cables with cable tie.

    Remarks : Cable Routing Criteria -

    - Individual separation layout
    - Cable do not get caught each other in the shelf parts
    - Masking Tape MUST remove

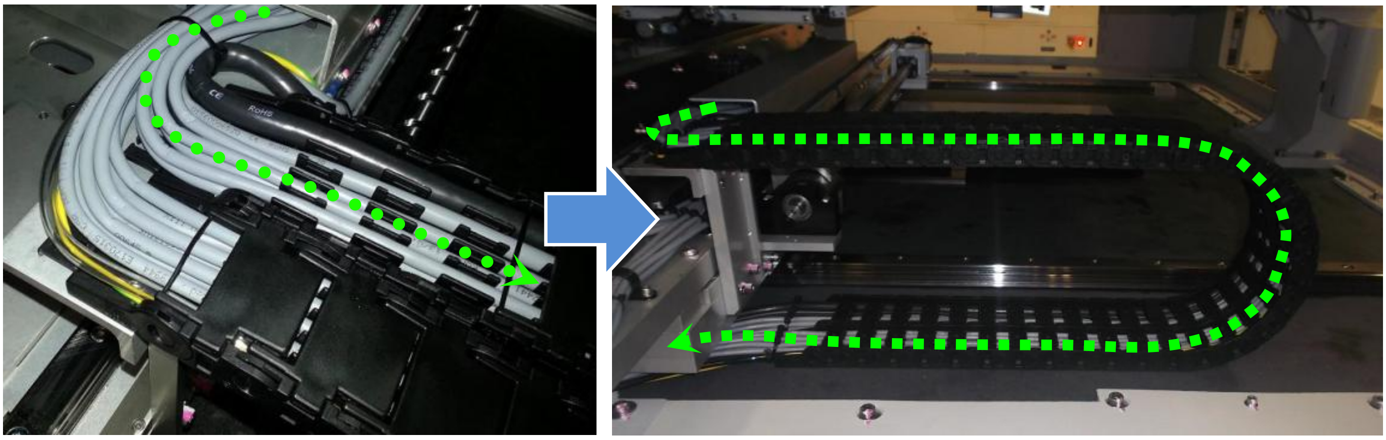


Figure 42 Stepper Motor Cable (Y axis Wide E-Chain)

45. Manually move Y axis stage towards Front & Rear rapidly and visual check Y axis E-chain where do not contain excessive bending radius.

Remarks : Re-route the cable if excessive bending radius addressed

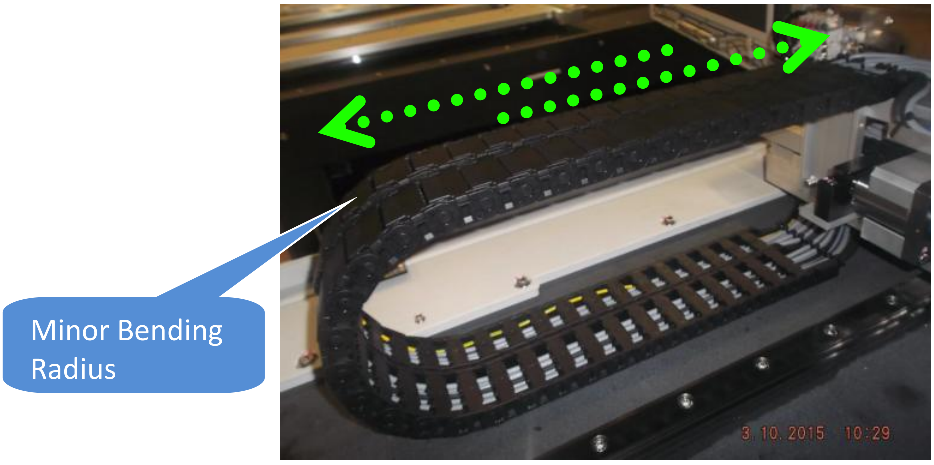


Figure 43 Stepper Motor Cable Routing (Y Axis Wide E-Chain)

46. Route 4JASC-A078 as shown and secure with necessary cable ties.

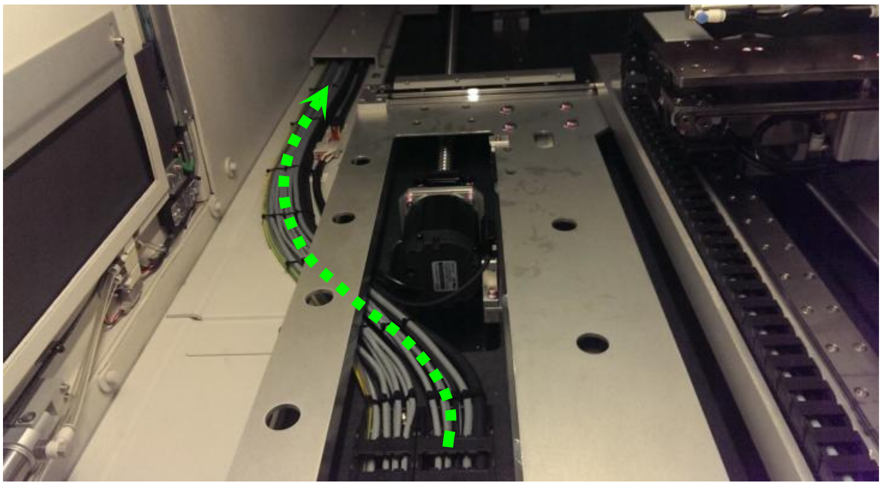


Figure 44 Stepper Motor Cable Routing (Y Axis Wide E-Chain)

47. Gently route 4JASC-A078 through the "Opening" as below Figure 45

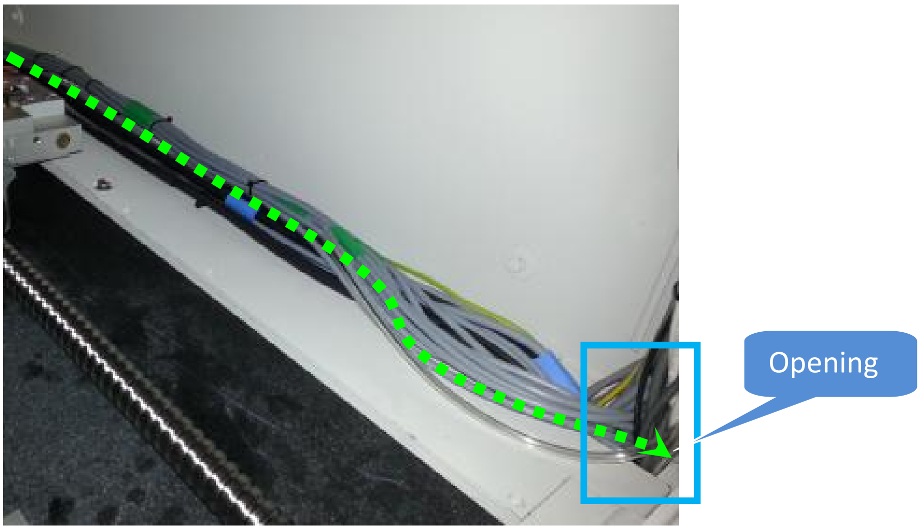


Figure 45

48. Access from rear side,assemble Vertical Lead Shielding Assembly by tighten 2 x screws.Next, assemble Cable Exit Shielding Assembly by tighten 3 x Screws.

Remarks :

- Caution on cable handling upon shielding assembly
- Gaping between both shielding assembly is not allow

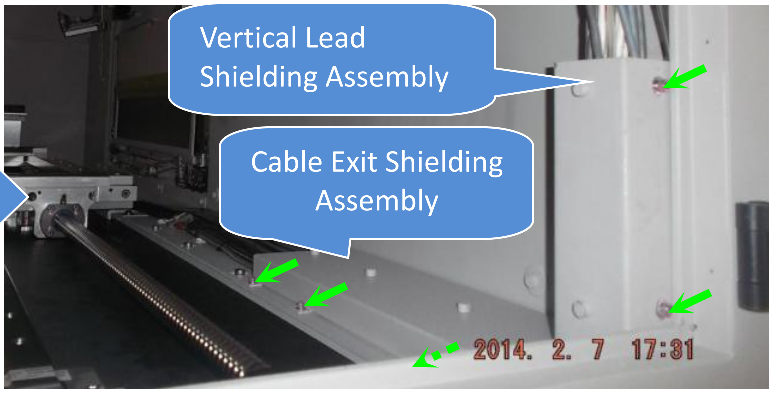


Figure 46 Shielding assembly

49. At Fix Rail, assemble PIP Cable Cover (P/N, 4JAXYS-N104) as below Figure 47

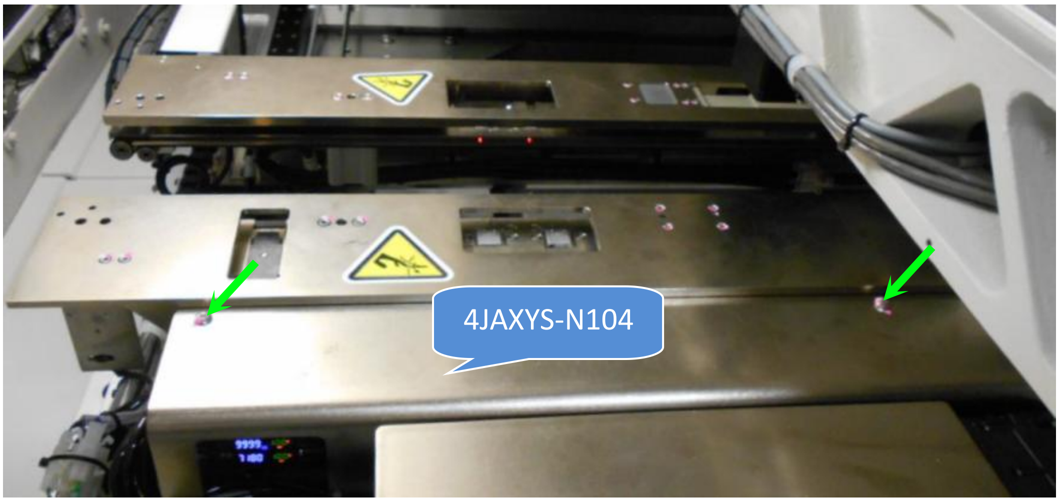


Figure 47

50. Route 4JASC-A078 as shown and apply necessary cable ties

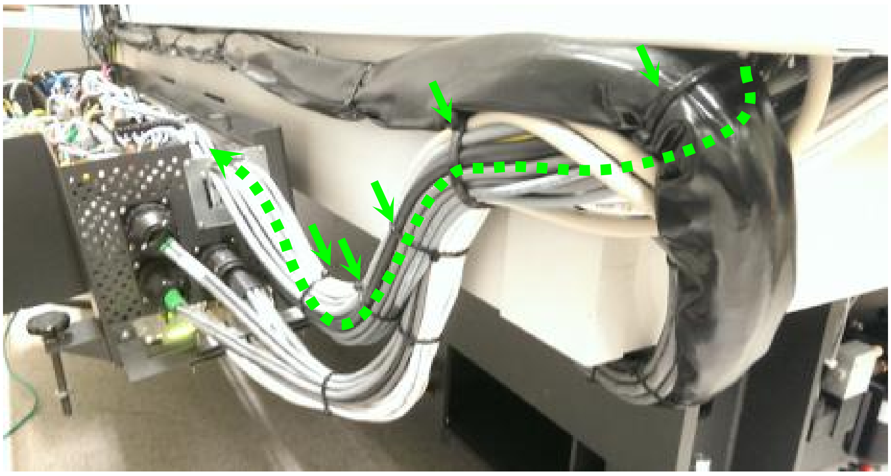


Figure 48 Stepper Motor Cable Routing (SSCA)

51. Route 4JASC-A078 as shown

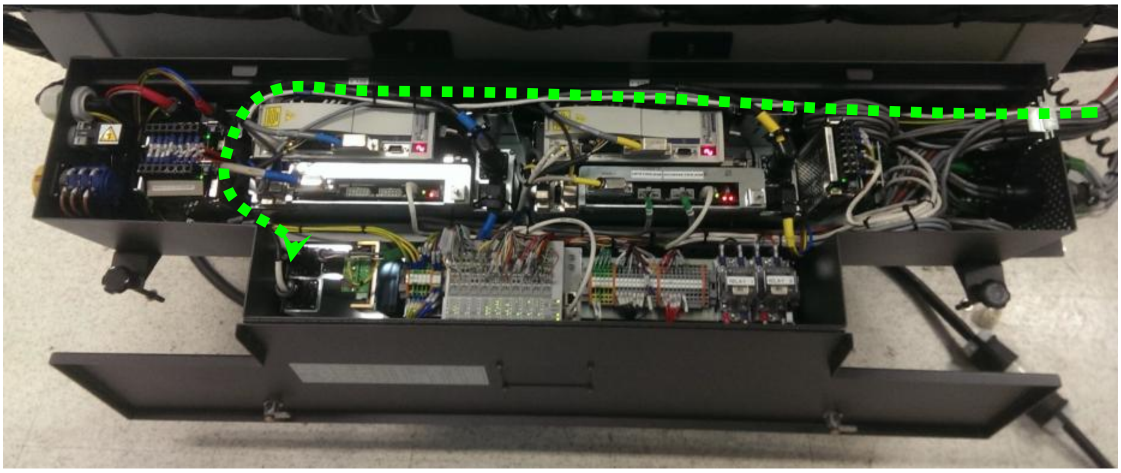


Figure 49 Stepper Motor Cable Routing (SSCA)

52. Connect end bulb head connector to AWA stepper driver

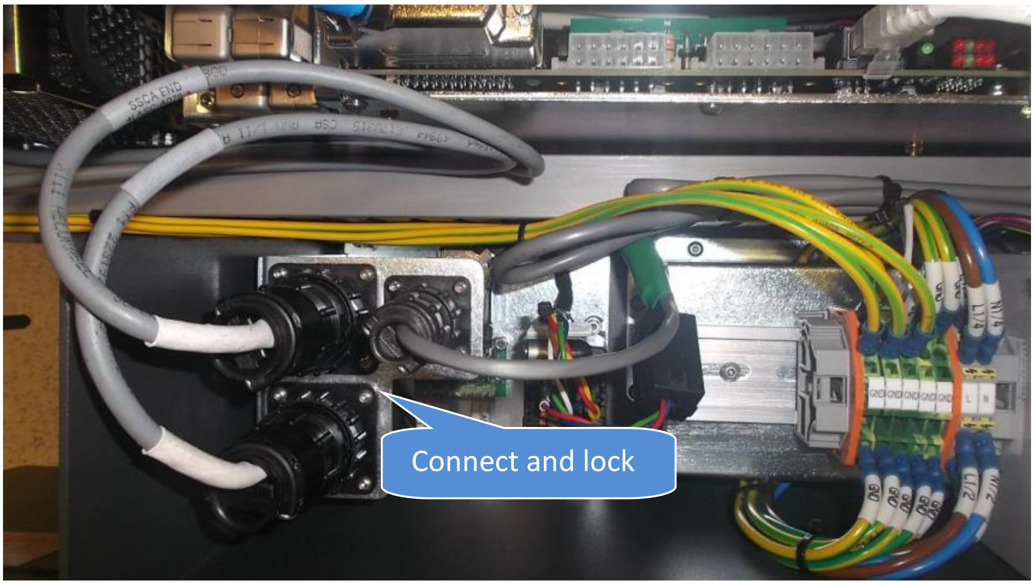


Figure 50 Stepper Motor Bulk Head Connector

53. Reassemble 4XAMC-A045 as shown. Tighten 2 x Screws accordingly

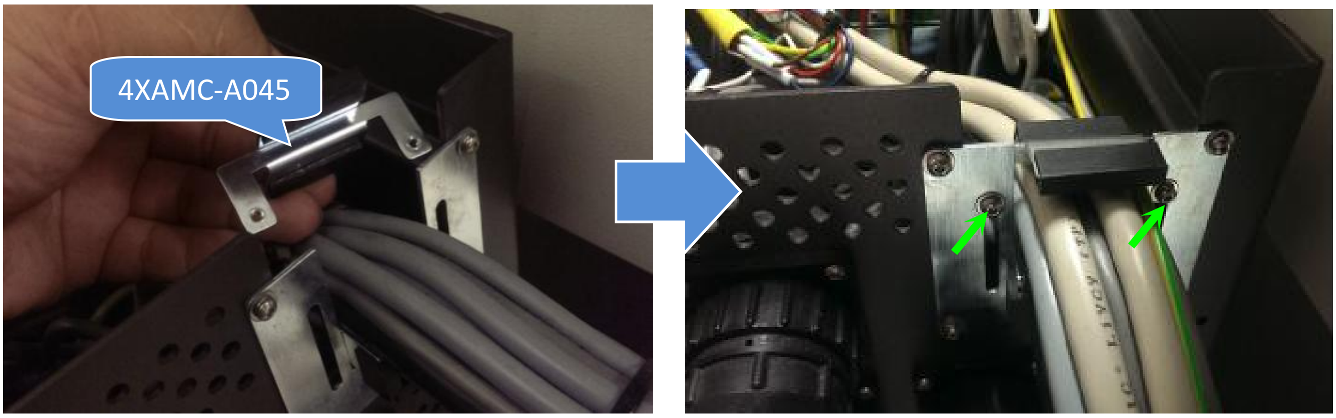


Figure 51 Holding Plate Assembly

54. Gently close outer barrier.Tighten up 2 x Screws.

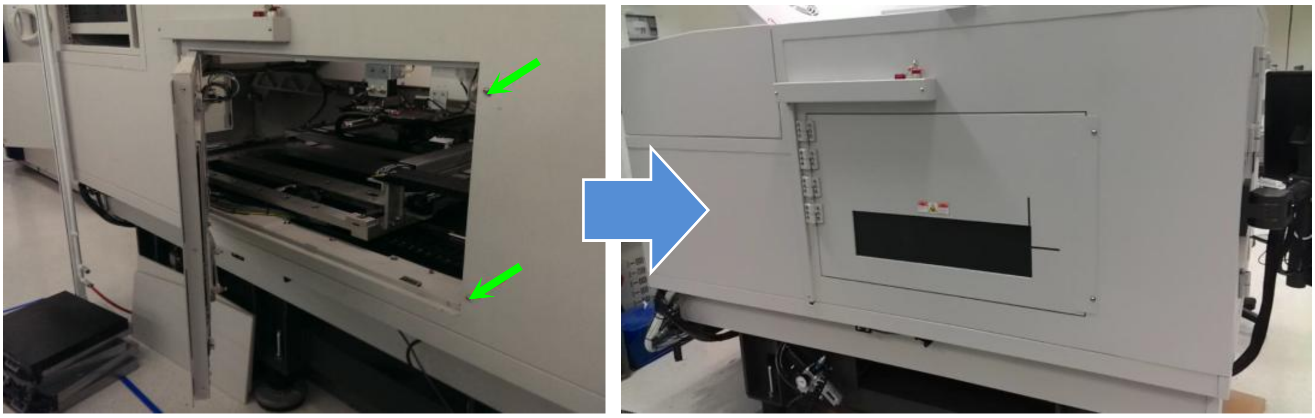


Figure 52

55. Turn On SSCA

56. Perform X ray Survey

AWA Rail Width Validation

57. Close all door access.Start-up machine as usual.

58. Log in V810 GUI and initialize DIO.

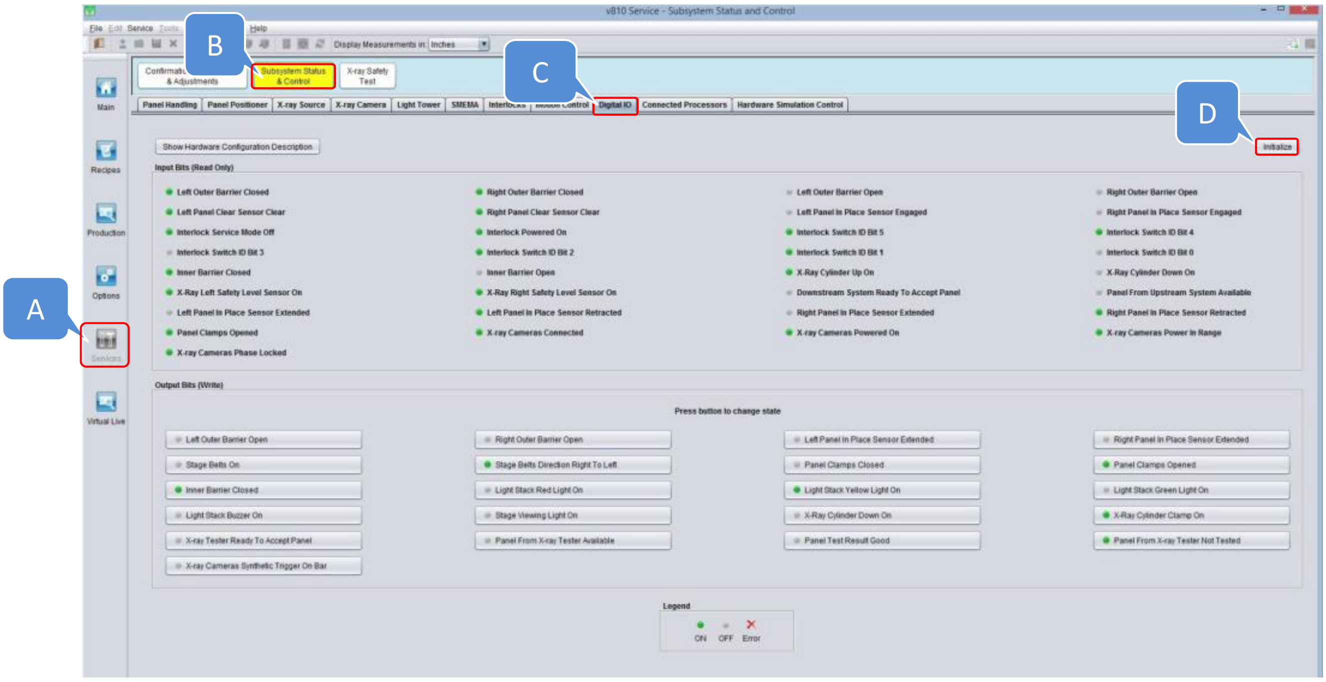


Figure 53 Digital IO Tab

59. Initialize at Panel Handling

60. Set the Display Measurement to millimeters as below Figure 54

61. Go to the Set Manual Adjustment as key in 350 and click “Set” as below Figure 54

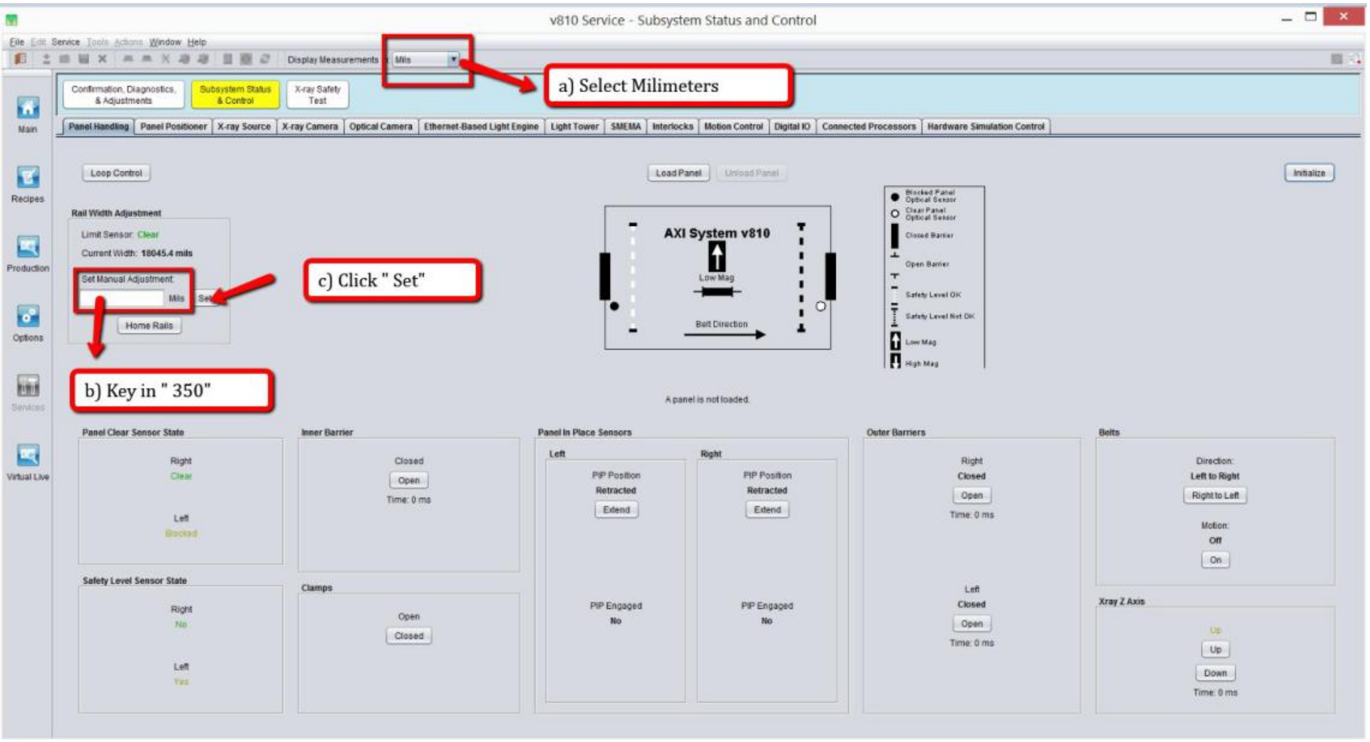


Figure 54 Panel Handling Interface

62. Wait until AWA is extend to desire width.

63. Close X-ray at Operator Console Panel and open the Front Door.

64. In advance, measure entire length of caliper.

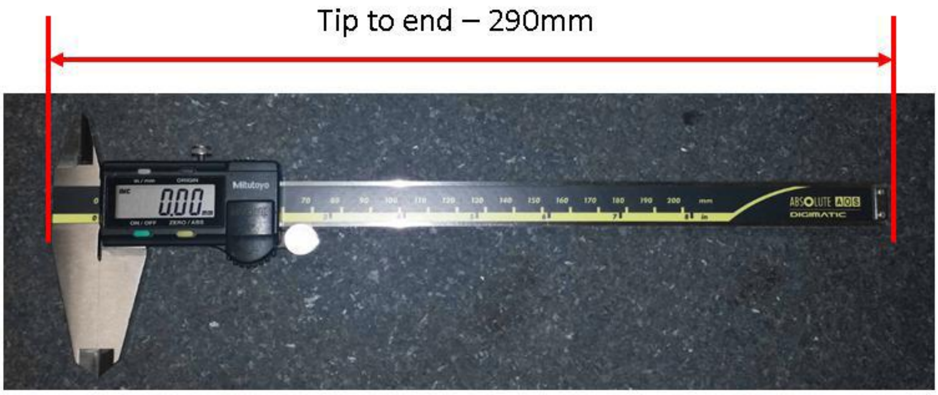


Figure 55 Caliper Length (Reference Pic Only)

65. Place Caliper as below position. (May refer to below Figure 56)

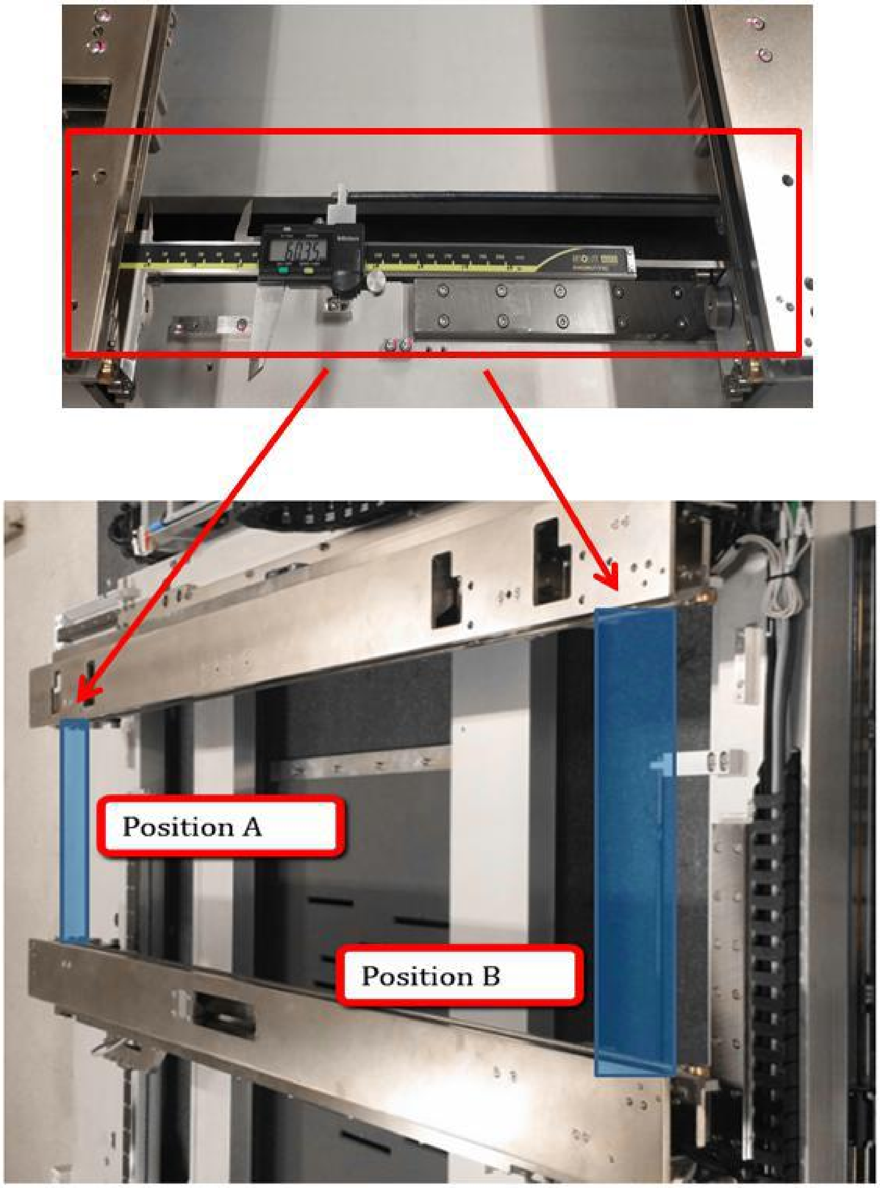
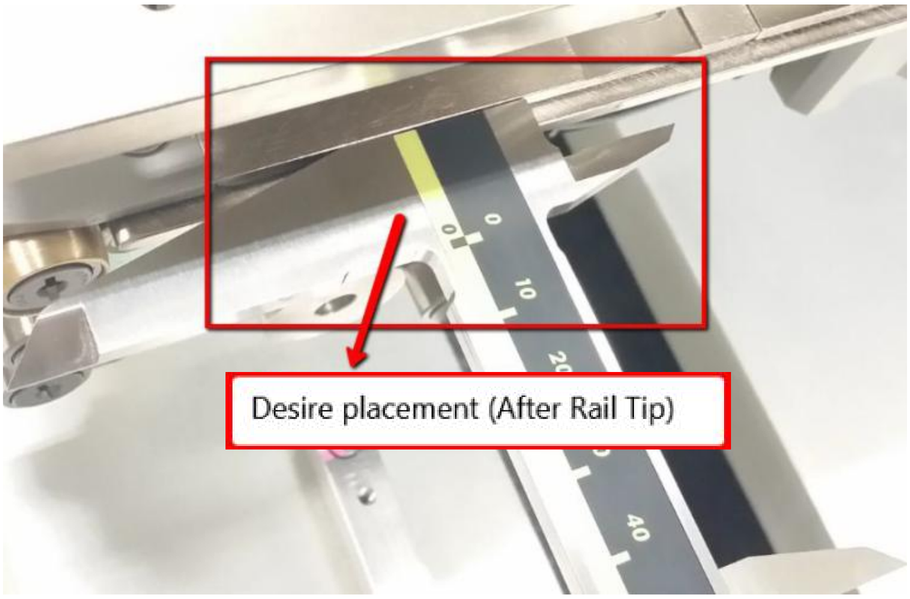
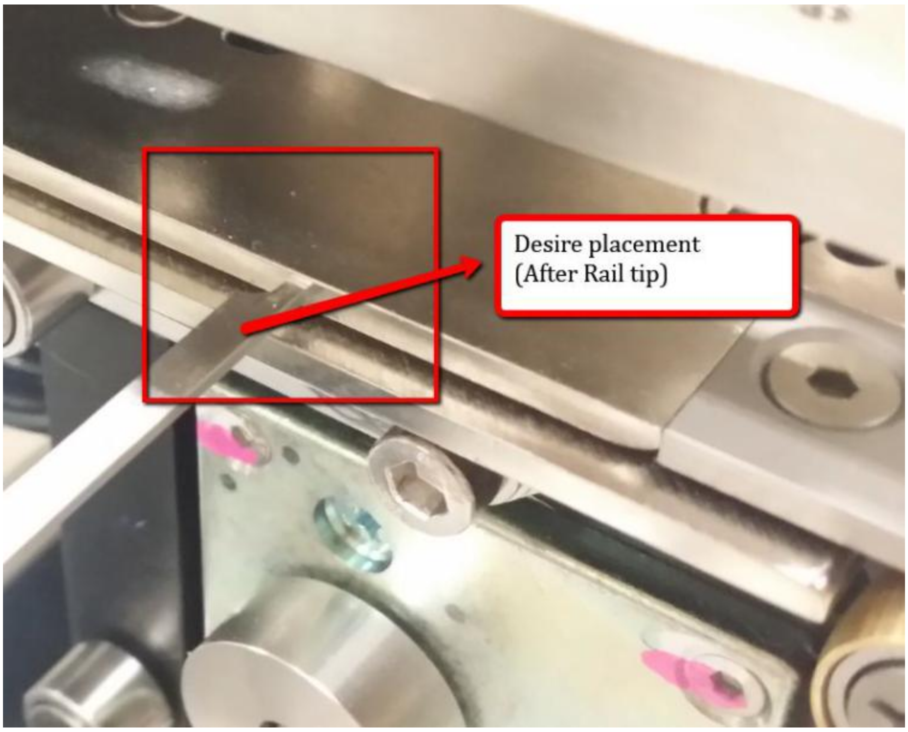
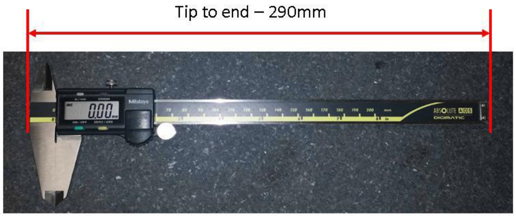
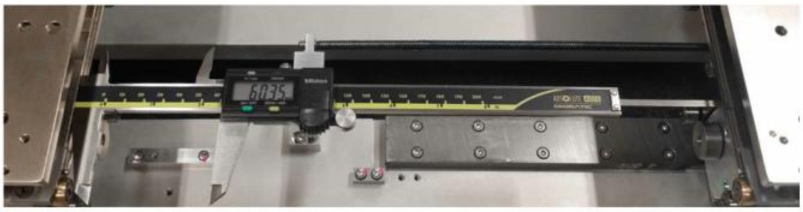


Figure 56 Caliper Placement

Position	Position A & Position B
Fix rail Caliper placement	
Adj rail Caliper placement	

66. Add up entire Caliper Length (Figure 55) with exact width of AWA

Item	Picture	Measurement
Exact length of Caliper		290
Exact Width of AWA		60.35mm
	Total	350.35

67. Record 5 repeated measurements at the right end (Position A), and then 5 repeated measurements at the left end (Position B):
    - MRave = average value of 5 repeated measurements at right end of conveyor
    - MLave = average value of 5 repeated measurements at left end of conveyor
68. Calculate the parallelism, Mave and width error using the formulas below. Compare these values to the parallelism and width error limits.
    - Parallelism = MRave - MLave
    - Mave = (MRave + MLave)/2
    - Width Error = Set Width - Mave
    - Limits on Parallelism = +/- 0.125mm
    - Limits on Width Error = +/- 0.125mm

69. By refer to below table and step, check Parallelism Limit & Width Accuracy Limit

Sample Measurement (mm)			
Measurement Right End (Position A)		Measurement Left End (Position B)	
MR1	350.00	ML1	350.10
MR2	350.01	ML2	350.08
MR3	350.05	ML3	350.08
MR4	350.00	ML4	350.06
MR5	350.01	ML5	350.09

Calculations:

Measurement Right End average (MRave) = (MR1+MR2+MR3+MR4+MR5)/5 = 350.01

Measurement Left End average (MLave)= (ML1+ML2+ML3+ML4+ML5)/5 = 350.08

Measurement average = (MRave + MLave)/2 = 350.04

Compare to Limits for Parallelism:

MRave – MLave = **0.07 is less than Spec "+/- 0.125mm ",** so no adjustment for parallelism is required

Compare to Limits for Width Accuracy:

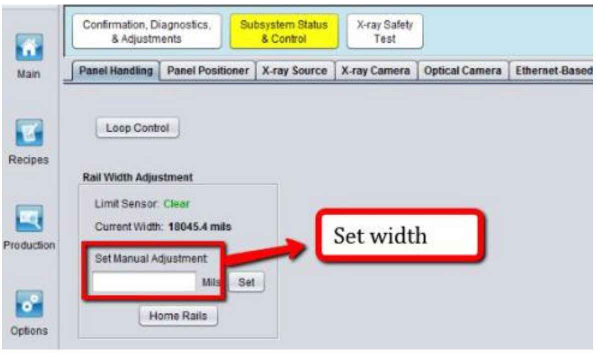


Figure 57 Set Width

Set Width – Mave = **0.04 is less than Spec "+/- 0.125mm ",** so no adjustment required for width accuracy

70. If not within spec, may refer to AXI V810 S2EX AWA Rail Width Adjustment Procedure for optimization.